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Analyses of Topical Policy Issues

Does common institutional ownership affect systemic risk of non-financial firms? Evidence from China

Xiao-Lin Li [a], Jiawei Hao [a], Liang Liu [b,*]

[a] School of Economics, Ocean University of China, Qingdao, Shandong, China
[b] Business School & Soochow University Think Tank, Soochow University, Suzhou, Jiangsu, China

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ABSTRACT

Using data on common institutional ownership (CIO) and systemic risk of Chinese listed manufacturing firms from 2010 to 2022, our paper investigates the effect of CIO on systemic risk of non-financial firms (NFFs). We find that CIO is positively associated with the systemic risk of NFFs. We identify two channels through which CIO influences NFFs' systemic risk: exacerbating risk accumulation through firm manipulation and increasing risk interconnectedness among firms. Our results are validated by several robustness tests. Heterogeneity analysis indicates that the effect of CIO on systemic risk is more pronounced in NFFs with lower stock liquidity, lower transparency, lower product market competition, and higher financing constraints. Finally, we find that CIO networks propagate individual firms' tail risk and thereby form a systemic risk danger. Our paper presents policy recommendations for regulators, stressing the need to effectively manage the development of CIO to mitigate its potential adverse effects, especially amid the Chinese government's promotion of institutional investors in capital markets.

1. Introduction

CIO refers to the equity correlation phenomenon in which an institutional investor holds shares in at least two same-industry firms, typically at a minimum shareholding threshold of 5 %. In recent years, driven by portfolio diversification and frequent mergers and acquisitions, the simultaneous ownership of multiple same-industry listed companies by institutional investors has become increasingly prevalent. Fig. 1(a) depicts the number of Chinese listed firms with CIO between 2010 and 2022, showing a clear upward trend throughout the sample period, with a particularly sharp increase in recent years. In the United States, about 60 % of listed companies are connected through CIO (He et al., 2017). Our data indicate that as of 2022, over 13 % of Chinese listed firms have CIO. As CIO becomes more widespread, interconnections among same-industry firms have strengthened. Fig. 1(b) illustrates an increasing number of firms connected via CIO within the same industry in China. Currently, academics remain divided regarding the impacts of CIO. Some studies argue that compared to individual investors, common institutional investors possess extensive industry knowledge and management expertise, motivating them to leverage synergistic governance to reduce excessive competition and enhance cooperation among their holdings. They also engage in corporate decision-making and oversight to mitigate principal-agent conflicts (e.g., Park et al., 2019; Chen et al., 2021). Conversely, other studies indicate that CIO may facilitate collusion among affiliated same-industry firms to enhance portfolio value, thereby diminishing market competition (e.g., Azar et al., 2018). Consequently, a debate persists over whether governments should impose stricter regulations on CIO. Our research contributes to this debate by exploring the effect of

* Corresponding author.
E-mail address: liul@suda.edu.cn (L. Liu).

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CIO on firm systemic risk.

Since financial crises often originate in the banking industry and related credit derivatives markets, most research concerning systemic risk focuses on financial firms (FFs). Extensive research has examined how internal determinants, including financial metrics, business frameworks, and governance mechanisms of financial institutions, influence systemic risk at the micro level (Anginer et al., 2018; Shim, 2019; Van Oordt and Zhou, 2019). At the macro level, studies have explored how external forces, such as macroeconomic conditions, financial market development, and government interventions, influence FFs' systemic risk (Matousek et al., 2020; Berger et al., 2020). Unlike most previous literature, our paper primarily examines NFFs to explore the drivers of systemic risk. NFFs can likewise hold systemic importance and may trigger systemic risk crises (Anginer et al., 2018). NFFs interact with the financial sector through multiple channels, such as participation in shadow banking, financial investments, and debt financing (Accornero et al., 2018; Si and Li, 2022). Additionally, NFFs establish significant links through trade credit, supply chains, production networks, and cross-shareholding by institutional investors (Li et al., 2023a). As a result, NFFs may hold systemic importance owing to their increasing interconnectedness with the financial system and one another. Therefore, studying NFFs' systemic risk is particularly important, given the Chinese government's focus on real economy development. Fig. 2 depicts the systemic risk of NFFs and FFs from 2010 to 2022, measured by $\Delta CoVaR$ and MES. In line with Dungey et al. (2022), these two measures assess the systemic contributions and vulnerabilities of individual firms. Throughout the study period, the systemic risk of NFFs strongly correlates with that of FFs. However, during specific sub-periods, particularly from 2011 to 2012 and 2016 to 2017, NFFs' systemic vulnerability exceeds that of FFs, as shown in Fig. 2(b).

Although existing research provides important insights into the drivers of systemic risk, the role of CIO in contributing to systemic risk remains insufficiently examined. Our study aims to expand research on systemic risk determinants by incorporating CIO as a key factor. Does CIO increase the systemic risk of NFFs? If so, what mechanisms drive this effect? Moreover, does this effect vary across firms with different characteristics? These questions remain underexplored in the academic literature.

Fig. 3 presents the systemic risk of manufacturing firms with and without CIO from 2010 to 2022, as measured by $\Delta CoVaR$ and MES. As a critical segment of NFFs, manufacturing firms constitute the backbone of the real economy. From 2010 to 2022, manufacturing firms with CIO exhibit notably higher systemic risk compared to those without CIO. Accordingly, we hypothesize that CIO increases NFFs' systemic risk.

To measure CIO, we collect institutional ownership data from the CSMAR database. We define the main explanatory variable as the natural logarithm of 1 plus the number of same-industry firms connected to the focal firm via its common institutional blockholder (CIB), where an institutional blockholder denotes an institutional investor holding at least 5 % of the company's shares. The main dependent variable is firm systemic risk. We employ the delta conditional value-at-risk ($\Delta CoVaR$) to measure NFFs' systemic risk. $\Delta CoVaR$ captures the extent to which the risk of individual NFF contributes to overall systemic risk as the economy transitions from normal conditions to a crisis state.

Based on an analysis of 22,515 firm-year observations from 2010 to 2022, covering 2505 listed Chinese manufacturing firms, we observe that CIO is positively correlated with NFFs' systemic risk. Exacerbating risk accumulation through firm manipulation and increasing risk interconnectedness among firms are two seemingly plausible channels. On the one hand, we find that CIBs manipulate company management to engage in leverage manipulation and earnings management, concealing firms' true leverage levels and profitability. This exacerbates information asymmetry and systemic risk accumulation. On the other hand, CIO enhances stock price synchronicity and expands firm trade credit, thereby strengthening risk interconnectedness and increasing the capacity for transmission and amplification of systemic risk.

To address endogeneity concerns, we utilize four methods: instrumental variable approach, propensity score matching (PSM) analysis, Heckman two-step model, and difference-in-differences analysis. Our findings remain robust to these endogeneity checks. Other robustness checks, including alternative measurements for the main variables, alternative institutional blockholder thresholds, controlling for industry and regional differences, and controlling for other shocks, confirm that the baseline regression findings remain statistically significant.

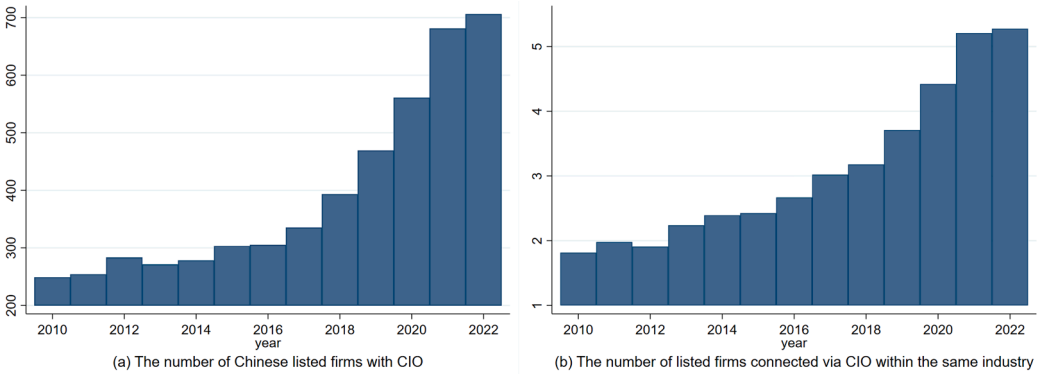

Fig. 1. The number of Chinese listed firms with CIO (panel a) and the number of listed firms connected via CIO within the same industry (panel b) between 2010 and 2022 in China.

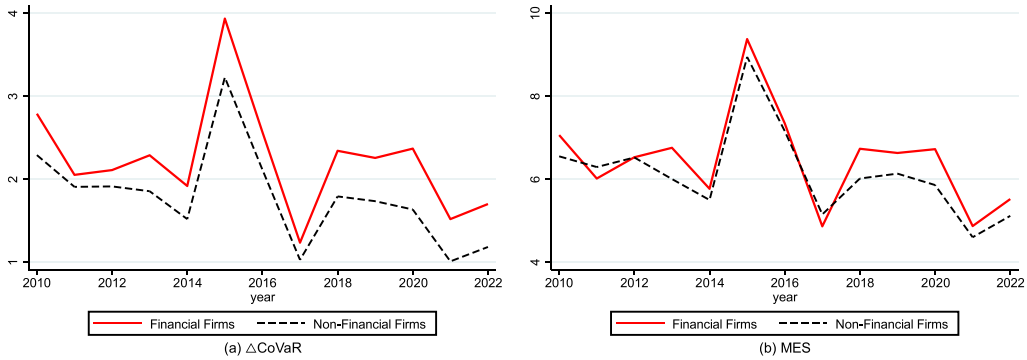

Fig. 2. Systemic risk of Chinese listed financial firms and non-financial firms from 2010 to 2022. Note: Systemic risk is measured by *ΔCoVaR* (panel a) proposed by Adrian and Brunnermeier (2016) and *MES* (panel b) introduced by Acharya et al. (2017).

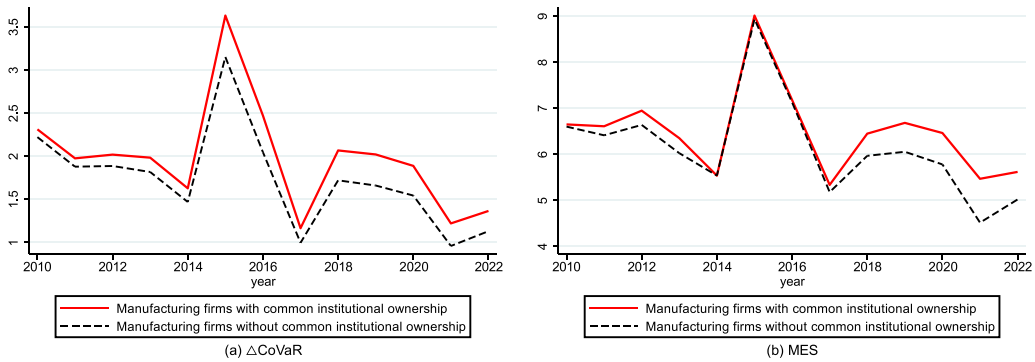

Fig. 3. Systemic risk of Chinese listed manufacturing firms with and without CIO from 2010 to 2022. Note: Systemic risk is assessed using *ΔCoVaR* (panel a) and *MES* (panel b).

In the heterogeneity analysis, we document that the positive association between CIO and systemic risk is more significant in NFFs with lower stock liquidity, lower transparency, lower product market competition, and higher financing constraints. Finally, narrowing our focus to a sample of NFFs with CIO, we find that CIO networks facilitate the transmission of individual firms' tail risk, thereby contributing to systemic risk formation.

Our paper's marginal contributions are as follows. First, we analyze the systemic risk determinants through the lens of CIO. While prior research has extensively explored various factors influencing firm systemic risk, the role of CIO remains underexplored. Even among the limited studies investigating CIO's impact on firm systemic risk (e.g., De George et al., 2017), CIO is primarily regarded as a linkage mechanism rather than a direct contributor to systemic risk accumulation within firms. Our study contributes to the literature by examining firm systemic risk through the lens of CIO, revealing that CIO not only strengthens inter-firm connectedness but also exacerbates systemic risk accumulation among NFFs.

Second, previous studies on systemic risk have predominantly focused on FFs, neglecting the potential systemic importance of NFFs (Dungey et al., 2022). Hence, our study redirects the focus toward manufacturing firms within the non-financial sector. Manufacturing firms are a major component of NFFs, serving as the backbone of the real economy. Investigating manufacturing firms' systemic risk is particularly important, especially given China's accelerated manufacturing transformation and upgrading, along with its efforts to prevent and mitigate systemic risks.

Third, our research offers valuable insights for regulators to standardize and guide CIO's development while mitigating firm systemic risk. As institutional investors expand their presence in China, the practice of holding stakes in multiple same-industry firms, leading to increased interconnections, has become increasingly prevalent. Some scholars and regulatory bodies have advocated for restrictions on CIO to curb monopolistic behavior (Lewellen and Lowry, 2021). According to the findings of our study, CIO increases NFFs' systemic risk. Therefore, given the Chinese government's promotion of institutional investors in capital markets, appropriately guiding CIO's development while mitigating its potential negative impacts is crucial for balancing stable growth and risk prevention.

This paper is structured as follows. Section 2 describes the institutional context and formulates the hypotheses. Section 3 outlines the estimation strategy. Section 4 details the primary empirical results. Section 5 offers a further analysis. Lastly, Section 6 concludes.

2. Institutional background and research hypothesis

2.1. Institutional background

In 1998, the launch of China's first closed-end fund marked the formal emergence of institutional investors in China. In 2000, the China Securities Regulatory Commission (CSRC) advocated for institutional investors' extraordinary development to accelerate the market-oriented reform of the fund industry. In 2001, the launch of China's open-end funds initiated an era of fund product differentiation and diversification of investment strategies. Subsequently, institutional investors gradually entered the capital market, increasing their influence. To further open the capital market, China's government introduced various key programs, such as the Qualified Foreign Institutional Investor scheme in 2002 and the Shanghai-Hong Kong Stock Connect in 2014. These policies allowed both foreign and mainland Chinese investors to trade eligible stocks across markets, facilitating relatively free capital flows in and out of China's stock market. Foreign capital inflows expanded the funding sources for Chinese institutional investors, propelling their development. Moreover, the advanced investment practices of foreign institutional investors pressured domestic counterparts to enhance their professionalism. Driven by portfolio diversification and frequent mergers and acquisitions, institutional investors increasingly hold shares of several same-industry firms simultaneously. As shown in Fig. 1, in recent years, the number of firms with CIO has steadily increased, and the interconnections among same-industry firms have been strengthened.

At present, China's economic recovery remains sluggish. Domestically, the sluggish real estate recovery and severe local government debt problems are intertwined with weak economic growth expectations. Externally, the Russia-Ukraine and Israel-Palestine conflicts, along with ongoing strategic competition between China and the U.S., place significant pressure on Chinese capital market development. China's government emphasized the need for continuous and effective risk prevention in key areas, underscoring the importance of resolutely safeguarding against systemic risks. On April 12, 2024, China released the new guiding opinions on the Capital Market, marking the third such set of guidance on promoting the high-quality development of capital market in the past two decades, following those in 2004 and 2014. The guiding opinions call for establishing a secure, transparent, and open capital market, emphasizing the need for enhanced regulation, risk prevention, and high-quality development. The guidance repeatedly emphasizes supervision, particularly at the corporate level, calling for stronger information disclosure and corporate governance oversight to enhance capital market stability. Therefore, based on the increasing phenomenon of CIO in China's capital market, studying its impact on NFFs' systemic risk and underlying mechanisms holds important guiding implications.

2.2. Research hypothesis

Systemic risk consists of two key dimensions: individual vulnerability and systemic interconnectedness (Van Oordt and Zhou, 2019). On the one hand, CIBs have incentives to manipulate company management for leverage manipulation and earnings management to maintain their information advantage for excess returns. These behaviors conceal a firm's true profitability and leverage level, exacerbating information asymmetry and leading to the accumulation of internal vulnerabilities in NFFs. On the other hand, CIO increases stock price synchronicity and corporate trade credit, coupled with the deepening integration of supply and industrial chains, has significantly enhanced systemic interconnectedness among NFFs in China. This heightened interconnectedness increases NFFs' exposure to un-diversifiable common risks. If a firm within the network encounters adverse shocks, it may trigger a "domino effect" of risk contagion, ultimately resulting in a systemic crisis (Benoit et al., 2017).

2.2.1. The manipulation collusion mechanism

To establish informational barriers to industry peers outside the holding companies and to other types of investors within the same firm (particularly minority shareholders) to pursue excess returns, CIBs have strong incentives to manipulate their holding companies to engage in leverage manipulation and earnings management. This practice further exacerbates information asymmetry and corporate risk vulnerability accumulation. On the one hand, CIBs typically seek to maximize portfolio value. In pursuit of this goal, they face fierce competition among portfolio firms as well as conflicts of interest and competitive pressures from non-shareholding peer firms. Conflicts within the portfolio are relatively manageable, whereas conflicts with external industry peers tend to be more intractable. Consequently, to confront these common external adversaries, shareholding firms have incentives to establish strategic alliances under CIO's guidance (Azar et al., 2018). Through earnings management and leverage manipulation, shareholding firms create informational barriers against external market participants by fabricating the appearance of low leverage, strong solvency, and robust profitability, thereby concealing their true earnings and leverage. This practice intensifies market information asymmetry and distorts the investment decisions of non-shareholding peer firms, ultimately diminishing industry competition. Consequently, the potential damage from aggressive competition among portfolio companies is mitigated, and firms within the strategic alliance may emerge as industry leaders, enabling CIBs to capture excess profits.

On the other hand, CIBs gather extensive firm-specific information and sector-specific expertise, serving as a bridge for information transmission from shareholding companies (Wang et al., 2023; Kang et al., 2018). Consequently, compared to other types of investors (particularly minority shareholders), CIBs have broader access to firm-specific information, which enables them to make more effective investment decisions and achieve superior returns over general investors in the capital market. To sustain this informational advantage, they tend to manipulate firms to engage in earnings management and leverage manipulation, thereby degrading the quality of information disclosures and constructing informational barriers. These barriers disrupt minority shareholders' trading strategies and exacerbate information asymmetry (Maffett, 2012; Ramalingegowda et al., 2021).

Leverage manipulation involves debt-like nominal equity, intentional use of off-balance-sheet liabilities, and various accounting

techniques to conceal actual debt, thereby reducing reported leverage levels (Zhu et al., 2024). By manipulating leverage, firms obscure their true financial risk, creating a false impression of deleveraging. Consequently, the actual leverage and financial stress far surpass the reported levels, resulting in the accumulation of hidden debt risks, heightened corporate vulnerability, and potential threats to macroeconomic stability (DeAngelo et al., 2018; Zhu et al., 2024). Specifically, from a micro perspective, leverage manipulation conceals excessive leverage and debt risks, creating the illusion of a financially stable firm. This misleading signal prompts external shareholders to expand investments, while creditors loosen loan covenants, further increasing the firm's actual leverage and exacerbating its risk vulnerability. From a macro perspective, while governmental decisions typically rely on macro indicators, many of these metrics stem from micro-level data. The low-leverage illusion induced by leverage manipulation may mislead regulators, resulting in lax supervision of high-risk firms or even misjudgment of their debt risks, thereby facilitating risk accumulation. When external shocks occur, firms with artificially low reported leverage are more susceptible to funding disruptions, increasing the probability of firm tail risk incidents (DeAngelo et al., 2018).

Earnings management is a key tool for managers to conceal unfavorable information from shareholders and investors. When managers employ earnings management to hide adverse internal information, financial statements no longer accurately reflect the firm's true economic performance, thereby exacerbating information asymmetry for external investors and increasing corporate vulnerability. Since earnings management incurs costs, firms weigh the expected benefits of concealing negative information against these costs. When the expected benefits fall short of the costs, managers may be compelled to disclose all previously hidden information simultaneously. This sudden revelation can prompt external investors to sell their shares, leading to a sharp fall in the firm's share price, thus exacerbating systemic risk (Jin and Myers, 2006).

2.2.2. The risk interconnectedness mechanism

CIO is pivotal in strengthening the interconnectedness among NFFs. On the one hand, it serves as equity linkages among same-industry NFFs, directly enhancing their interconnectedness. On the other hand, the increased stock price synchronicity and expanded trade credit induced by CIO indirectly deepen the interconnectedness among NFFs. This increased interconnectedness strengthens the transmission and amplification of systemic risk. Consequently, when a firm experiences a shock, the risk rapidly propagates to interconnected companies across the economic system, creating a "domino effect" of risk contagion that may trigger a systemic crisis (Benoit et al., 2017).

When institutional investors enter or exit a certain industry, or increase or decrease their holdings, their capital flow can result in stock price co-movement among connected firms (Gao et al., 2017). Stock price synchronicity reflects the relative quantity of firm-specific information influencing stock prices (Haghighat et al., 2015). Less firm-specific information in stock prices results in higher stock price synchronicity. CIBs generally possess much information due to their expertise and close interaction with management. However, they tend to engage in collusive practices to reduce the firm-specific information in stock prices, thereby exacerbating information asymmetry for external investors. Consequently, external investors acquire relatively less firm-specific information, leading to higher stock price synchronicity in firms with CIO. Higher stock price synchronicity indicates stronger co-movement with the broader capital market, implying that individual stock prices primarily reflect overall market and industry fluctuations rather than firm fundamentals. As a result, external investors struggle to accurately assess a firm's true financial condition and risk exposure. During negative shocks, external investors sell off their holdings to mitigate exposure, driven by their inability to distinguish firm-specific risks. This widespread sell-off amplifies market volatility and increases the likelihood of systemic risk events (An and Zhang, 2013). In practice, financial crises often coincide with widespread stock price crashes, and systemic risk accumulation preceding a crisis is typically marked by many stocks reaching price ceilings. These extreme phenomena of stock price co-movement contribute to the outbreak of systemic risk events.

CIBs typically aim to maximize their portfolio value. In pursuit of this goal, they coordinate competition and resolve conflicts among shareholding peer firms, thereby fostering strategic alliances among CIO peer firms (Azar et al., 2018). These alliances enhance the bargaining power and market influence of shareholding firms, enabling them to obtain more trade credit in transactions with upstream and downstream partners. Liu and Hou (2023) demonstrate that firms with CIO gain more trade credit mainly due to the anticompetitive effect. Trade credit, a type of short-term debt financing, reflects the connections between non-banking lenders and borrowers. Trade credit strengthens the risk interconnections among firms (Dungey et al., 2022). Moreover, firms with more trade credit perform poorly during financial crises due to the increased debt pressure. NFFs' trade credit practices enhance the credit linkages between upstream and downstream firms within the supply chain and across different industrial chains, creating a complex trade credit network. These linkages facilitate the transfer of default risk from a single firm to its upstream and downstream partners through the "balance sheet contagion" effect, potentially propagating further across the supply chain network (Kiyotaki and Moore, 2002). This accumulation of risk and spillover effects can trigger systemic risk within the economic system. Therefore, the increased trade credit driven by CIO can be viewed as a potential risk realization process for systemic risk in NFFs.

According to our preceding analyses, we put forward the following hypotheses:

Hypothesis 1. CIO increases NFFs' systemic risk.

Hypothesis 2. CIO increases NFFs' systemic risk through two key channels: the manipulation collusion mechanism and the risk interconnectedness mechanism.

Nonetheless, some factors suggest that our proposed relationship may not hold, or that CIO could decrease firm systemic risk. The collaborative governance effect of CIO may reduce the risks faced by firms, alleviating the accumulation of risk factors (Ni and Yin, 2023; Chen et al., 2024). Studies show that CIO can reduce systemic risk by improving corporate governance, reducing information asymmetry, internalizing governance externalities, and raising awareness of firm social responsibility. Within the CIO corporate governance framework, institutional investors offload distressed assets in response to liquidity shocks, thereby enhancing company

governance via exit and engagement mechanisms (Edmans et al., 2019). CIO, with its information advantages, reduces corporate misconduct and strengthens supervision (Wang et al., 2023; Kang et al., 2018). Internalizing governance externalities enables CIO to incur lower information processing costs and possess stronger incentives to monitor corporate behavior, thereby mitigating firms' systemic risk (Chen et al., 2024; Ni and Yin, 2023). DesJardine et al. (2023) contend that CIO enhances awareness of firm social responsibility and generates spillover effects that reduce systemic risk.

3. Estimation strategy

3.1. Econometric specification

To identify the causal relationship between CIO and NFFs' systemic risk, we employ *LnNumCon*, representing the number of same-industry firms sharing a CIB with the focal firm, as the main explanatory variable, and *ΔCoVaR*, the change in conditional value-at-risk, as the main dependent variable. We establish the following baseline econometric specification:

$$\Delta CoVaR_{i,t} = \alpha_0 + \alpha_1 LnNumCon_{i,t-1} + \alpha_2 Controls_{i,t-1} + \mu_i + \lambda_t + \varepsilon_{i,t} \tag{1}$$

where i denotes a specific NFF, and t represents a year. *ΔCoVaR* measures NFFs' systemic risk. *LnNumCon* reflects the connectivity degree between the focal firm and its peers via CIO. *Controls* denote several control variables. We include year-fixed effect λ and individual-fixed effect μ to control temporal macroeconomic changes and firm-specific heterogeneity, respectively. Robust standard errors are clustered at the firm level. α_1 reflects the influence of CIO on NFFs' systemic risk. Building on our theoretical analysis, α_1 is expected to be positive. To mitigate endogeneity issues and account for the lagged effect of CIO on firm systemic risk, all independent variables in model (1) are lagged by one period.

3.2. Variable definitions

We employ *ΔCoVaR* to quantify NFFs' systemic risk, computed following Adrian and Brunnermeier (2016). Recent research confirms that *ΔCoVaR* is equally effective in evaluating NFFs' systemic risk (Dungey et al., 2022; Li et al., 2023b). Specifically, when NFFs encounter external shocks such as supply chain disruptions, abrupt shifts in market demand, or regulatory interventions, these events can trigger the propagation of both internal and external risks. In such cases, *ΔCoVaR* quantifies a firm's marginal contribution to the overall systemic risk under extreme conditions, thereby revealing potential risk spillovers among NFFs. Notably, compared to FFs, NFFs face more diverse risk sources and exhibit greater data dispersion and heterogeneity. Therefore, introducing *ΔCoVaR* into the risk assessment of NFFs can reflect the complex structure of NFFs' systemic risk more comprehensively.

$\Delta CoVaR_t^{m|i}$ denotes the systemic contribution for firm i to market m at time t under extreme conditions. The calculation is expressed as follows:

$$\Delta CoVaR_t^{m|i} = CoVaR_t^{m|VaR_{1-q}^i} - CoVaR_t^{m|VaR_{50\%}^i} \tag{2}$$

where VaR_{1-q}^i denotes firm i's Value-at-Risk, representing the maximum potential loss at a confidence level of 1-q, with q set to 5 %. Conversely, $VaR_{50\%}^i$ reflects firm i's Value-at-Risk under normal market conditions at a 50 % confidence level. $CoVaR_t^{m|VaR_{1-q}^i}$ signifies the Value-at-Risk for market m when firm i is under distress, whereas $CoVaR_t^{m|VaR_{50\%}^i}$ signifies the Value-at-Risk for market m under normal conditions. Utilizing the DCC-GARCH model, we estimate the weekly *ΔCoVaR* at a 95 % confidence level based on the weekly return data from listed firms and then compute the annual average. To enhance the clarity of the results, we adopt the negative of *ΔCoVaR*, where larger values indicate increased risk contribution for a firm to the market, reflecting higher systemic risk.

To construct CIO measures, we derive institutional ownership data from the CSMAR database. If an investor holds a minimum of 5 % ownership in a company, it is classified as a blockholder (Cheng et al., 2022). Therefore, we define CIO as the ownership by institutional blockholders who simultaneously hold stakes in at least two same-industry firms. Given that the manufacturing industry constitutes a significant proportion of listed enterprises, we use the second-level industry codes issued by CSRC when calculating CIO indicators for this sector. The primary independent variable, *LnNumCon*, represents the number of same-industry firms linked to firm i via its CIB. We first compute quarterly values and then average them over four quarters to gain annual indicators. To mitigate skewness, we apply the natural logarithm of 1 plus the values.

In model (1), we include some control variables that may influence NFFs' systemic risk: (i) firm size (*Size*), measured as the natural logarithm of total assets; (ii) beta value (*Beta*), defined as the risk coefficient derived from regressing firm stock returns on market stock returns; (iii) listing age (*ListAge*), proxied by the natural logarithm of its listing age; (iv) leverage (*Lev*), quantified by the ratio of total liabilities to total assets; (v) growth opportunity (*TobinQ*), proxied by the ratio of a firm's market value to the replacement cost of its total assets; (vi) return on equity (*ROE*), determined as the percentage of net profit to shareholders' equity; (vii) CEO duality (*Dual*), a binary variable suggesting whether the CEO concurrently serves as the Chairman; (viii) independent director ratio (*Indep*), calculated as the percentage of independent directors to total number of directors; (ix) ownership concentration (*Top10*), calculated as the percentage of total shares owned by firm's top 10 shareholders; (x) management ownership (*Mshare*), measured as the proportion of shares owned by a firm's executives.

3.3. Sample selection

Our empirical analysis uses data on Chinese listed manufacturing firms from 2010 to 2022. Their annual financial data are sourced from the CSMAR database. We also obtain the stock return data required for risk indicator calculations from China's Wind database. The sample is processed as follows: (1) deleting observations with missing main variables; (2) removing firms identified as Special Treatment; and (3) deleting firms listed after 2020 to consider the possible delayed influence of CIO on NFFs' systemic risk. To reduce the influence of outliers, all continuous variables are winsorized at the 1 % and 99 % levels. After preprocessing, our final research sample comprises 2505 firms and 22,515 observations.

3.4. Summary statistics

Panel A of Table 1 presents descriptive statistics for our key variables. *ΔCoVaR* varies between 0.473 and 4.173, with an average of 1.751, suggesting substantial differences in NFFs' systemic risk. By comparing the means and standard deviations, we observe substantial differences in profitability and managerial ownership ratios among the sample firms. Panel B displays the univariate analysis of differences between NFFs with and without CIO. NFFs with CIO exhibit significantly higher systemic risk. The P-value (0.000) indicates this difference is significant. In addition, NFFs with CIO may have larger size, more growth opportunities, greater profitability, and higher leverage than NFFs without CIO. These findings preliminarily indicate a possible positive impact of CIO on NFFs' systemic risk.

4. Empirical results

4.1. Baseline regression results

Table 2 displays the findings regarding the impact of CIO on NFFs' systemic risk. Column (1) shows regression results controlling only for year-fixed and individual-fixed effects. α_1 is significantly positive, providing preliminary support for Hypothesis 1. From columns (2) to (5), we incrementally add firm-level control variables, to minimize omitted variable bias. α_1 remains significantly positive, further validating Hypothesis 1. Economically, after controlling for all control variables, as indicated by column (5), a one-standard-deviation (0.338) increase in *LnNumCon* results in a 0.014 (0.338 × 0.041) rise in NFFs' systemic risk. These findings indicate

Table 1
Descriptive statistics of the main variables.

Panel A: Descriptive Statistics

Variable	N	Mean	SD	Min	P50	Max
ΔCoVaR (%)	22,515	1.751	0.722	0.473	1.659	4.173
LnNumCon	22,515	0.095	0.338	0.000	0.000	2.048
Size	22,515	22.026	1.165	19.951	21.874	25.595
Beta	22,515	1.052	0.778	−1.381	1.042	3.848
ListAge	22,515	2.011	0.873	0.000	2.079	3.332
Lev (%)	22,515	0.392	0.192	0.051	0.384	0.861
TobinQ (%)	22,515	2.116	1.301	0.877	1.697	8.446
ROE (%)	22,515	0.067	0.124	−0.553	0.073	0.363
Dual	22,515	0.319	0.466	0.000	0.000	1.000
Indep (%)	22,515	0.376	0.053	0.333	0.333	0.571
Top10 (%)	22,515	0.580	0.148	0.230	0.587	0.887
Mshare (%)	22,515	15.855	20.710	0.000	2.678	69.959

Panel B: Univariate Analysis

	Non-CIO Firms		CIO Firms		Difference in mean	*P*-value of difference
	N	Mean	N	Mean		
ΔCoVaR (%)	20,348	1.736	2167	1.885	−0.148	0.000***
Size	20,348	21.919	2167	23.034	−1.115	0.000***
Beta	20,348	1.049	2167	1.084	−0.035	0.046**
ListAge	20,348	1.965	2167	2.444	−0.479	0.000***
Lev (%)	20,348	0.387	2167	0.444	−0.057	0.000***
TobinQ (%)	20,348	2.100	2167	2.265	−0.165	0.000***
ROE (%)	20,348	0.065	2167	0.089	−0.024	0.000***
Dual	20,348	0.328	2167	0.236	0.092	0.000***
Indep (%)	20,348	0.376	2167	0.377	−0.002	0.188
Top10 (%)	20,348	0.579	2167	0.595	−0.016	0.000***
Mshare (%)	20,348	16.804	2167	6.945	9.859	0.000***

Note: This table reports the descriptive statistics (Panel A) and univariate analysis of differences in main variables between NFFs without CIO and NFFs with CIO (Panel B). The sample consists of 22,515 firm-year observations from 2010 to 2022. *, **, and *** denote the significance levels of 10 %, 5 %, and 1 %, respectively.

Table 2
Impact of common institutional ownership on the systemic risk of non-financial firms.

Variables	(1) △*CoVaR*	(2) △*CoVaR*	(3) △*CoVaR*	(4) △*CoVaR*	(5) △*CoVaR*
LnNumCon	0.041***	0.030***	0.031***	0.032***	0.041***
	(3.757)	(2.793)	(2.923)	(2.998)	(3.737)
Size		0.096***	0.107***	0.103***	0.112***
		(12.101)	(12.548)	(11.939)	(13.005)
Beta		0.010***	0.010***	0.010***	0.010***
		(3.207)	(3.246)	(3.237)	(3.157)
ListAge		0.071***	0.099***	0.102***	0.060***
		(8.713)	(11.318)	(11.602)	(6.072)
Lev			−0.212***	−0.195***	−0.221***
			(−7.152)	(−6.351)	(−7.285)
TobinQ			−0.011***	−0.012***	−0.012***
			(−3.134)	(−3.521)	(−3.399)
ROE				0.079***	0.096***
				(2.643)	(3.222)
Dual				0.001	0.004
				(0.161)	(0.385)
Indep					−0.089
					(−1.085)
Top10					−0.420***
					(−9.258)
Mshare					−0.000
					(−0.993)
Constant	1.755***	−0.374**	−0.533***	−0.463**	−0.297
	(212.415)	(−2.222)	(−2.972)	(−2.563)	(−1.621)
IE	YES	YES	YES	YES	YES
YE	YES	YES	YES	YES	YES
Adj-R^2	0.732	0.737	0.738	0.739	0.741
N	19,353	19,353	19,353	19,353	19,353

Note: This table examines the impact of CIO on NFFs' systemic risk. *, **, and *** denote the significance levels of 10 %, 5 %, and 1 %, respectively. The values in parentheses represent t-statistics adjusted for robust standard errors. Here, *IE* refers to individual fixed effects, while *YE* denotes time fixed effects.

CIO increases NFFs' systemic risk.

We find that the control variables exhibit the anticipated signs. For instance, the positive coefficients for *Size* suggest that larger NFFs face greater systemic risk, likely reflecting the "too big to fail" risk. Additionally, the coefficients for *Beta* are notably positive, implying a positive relationship between *Beta* and systemic risk.

4.2. Potential mechanisms of the main results

4.2.1. Manipulation collusion mechanism

Based on the previous theoretical analyses, CIBs have incentives to manipulate company management for leverage manipulation and earnings management, thereby securing excess returns. These behaviors conceal the firm's true profitability and leverage level, exacerbating information asymmetry and leading to the accumulation of internal risk vulnerabilities in NFFs.

First, firms can manipulate leverage not only through on-balance sheet debt but also by off-balance sheet liabilities and disguised debt-equity instruments. We evaluate leverage manipulation as the difference between the actual leverage—accounting for disguised debt-equity financing and off-balance sheet liabilities—and the reported book leverage ratio. We calculate two indicators, *LEVM* and *ExpLEVM*, using the basic and extended XLT-LEVM methods introduced by Xu et al. (2020). *LEVM* captures leverage manipulation solely through disguised debt-equity financing and off-balance sheet liabilities, while *ExpLEVM* extends *LEVM* by incorporating accounting strategies such as fixed asset depreciation and R&D capitalization. Columns (1)-(2) of Table 3 display that CIO significantly enhances NFFs' leverage manipulation. Through leverage manipulation, firms obscure the true financial risk, creating an illusion of deleveraging. Consequently, actual leverage and financial stress exceed reported levels, leading to the accumulation of hidden debt risks and heightened risk vulnerability.

Second, we adopt the real earnings management (REM) approach proposed by Roychowdhury (2006). REM refers to deviations from normal operational practices, wherein managers manipulate reported earnings through real economic decisions (Roychowdhury, 2006). Compared to accrual earnings management, REM is generally a stronger predictor of firm-specific stock price crashes (Li et al., 2011). We obtain REM data from the CSMAR database. Column (3) of Table 3 indicates that CIO significantly enhances NFFs' REM. The findings suggest that CIBs manipulate managers to engage in earnings management, concealing adverse internal information. Consequently, the company's financial records fail to accurately represent its true economic performance, thereby exacerbating information asymmetry for external investors and increasing the firm's risk vulnerability. In conclusion, CIO significantly enhances leverage manipulation and earnings management in NFFs, thereby increasing their risk vulnerability. These findings partially support Hypothesis 2, validating the manipulation collusion mechanism through which CIO increases NFFs' systemic risk.

Table 3
Channel analysis: the manipulation collusion mechanism.

Variables	(1) Leverage manipulation	(2)	(3) Earnings management
	LEVM	*ExpLEVM*	*REM*
LnNumCon	0.006*	0.006*	0.011*
	(1.923)	(1.860)	(1.849)
Size	−0.034***	−0.033***	0.031***
	(−9.646)	(−9.520)	(5.310)
Beta	−0.001	−0.001	0.001
	(−0.589)	(−0.671)	(0.440)
ListAge	0.009**	0.009*	−0.013
	(1.985)	(1.900)	(−1.306)
Lev	0.006	0.010	0.078***
	(0.483)	(0.835)	(4.093)
TobinQ	0.004***	0.005***	−0.017***
	(4.151)	(4.202)	(−9.215)
ROE	0.080***	0.079***	−0.259***
	(8.209)	(8.138)	(−14.217)
Dual	−0.002	−0.002	−0.005
	(−0.771)	(−0.774)	(−0.996)
Indep	−0.011	−0.010	0.010
	(−0.443)	(−0.413)	(0.249)
Top10	−0.059***	−0.060***	0.018
	(−3.518)	(−3.551)	(0.621)
Mshare	−0.000**	−0.000**	−0.000
	(−2.260)	(−2.126)	(−0.688)
constant	0.820***	0.809***	−0.636***
	(11.083)	(10.958)	(−4.995)
IE	YES	YES	YES
YE	YES	YES	YES
Adj-R^2	0.048	0.047	0.078
N	18,933	18,933	19,899

Note: This table presents the results of the manipulation collusion mechanism through which CIO increases NFFs' systemic risk. *, **, and *** denote the significance levels of 10 %, 5 %, and 1 %, respectively. The values in parentheses represent t-statistics adjusted for robust standard errors. Here, *IE* refers to individual fixed effects, while *YE* denotes time fixed effects.

4.2.2. Risk interconnectedness mechanism

Prior theoretical analyses indicate that CIO enhances stock price synchronicity and expands trade credit, thereby elevating risk interconnectedness among firms and increasing the capacity for systemic risk transmission and amplification.

First, in line with Boubaker et al. (2014), we compute R^2 using the following regression specifications:

$$r_{i,t} = \alpha + \beta_1 r_{m,t} + \gamma_1 r_{l,t} + \varepsilon_{i,t} \tag{3}$$

$$r_{i,t} = \alpha + \beta_1 r_{m,t} + \beta_2 r_{m,t-1} + \gamma_1 r_{l,t} + \gamma_2 r_{l,t-1} + \varepsilon_{i,t} \tag{4}$$

where $r_{i,t}$ denotes the return of firm i during week t. $r_{m,t}$ is the market return, calculated using the value-weighted average method. $r_{l,t}$ is the return of industry l. Then we apply model (5) using R^2 to derive two stock price synchronicity measures, *SYN1* and *SYN2*.

$$SYN_{i,t} = \ln\left(\frac{R^2_{i,t}}{1 - R^2_{i,t}}\right) \tag{5}$$

Columns (1)-(2) of Table 4 indicate that CIO significantly increases NFFs' stock price synchronicity. It implies that CIBs manipulate corporate management to reduce the reliability of disclosed information. Consequently, the amount of firm-specific information in stock prices decreases, ultimately resulting in higher stock price synchronicity. Higher stock price synchronicity indicates a stronger co-movement between a company's share price and overall capital market, thereby expanding the risk exposure of firms with CIO and intensifying their risk interconnectedness with other firms. When faced with negative shocks, the stock prices of firms with CIO tend to decline in tandem with the overall market. As external investors struggle to distinguish firm-specific risks, they are more inclined to conduct widespread sell-offs to avoid risk, thereby exacerbating capital market volatility and heightening the likelihood of systemic risk events.

Second, following Liu and Hou (2023), we define the variable *TC* to measure firm trade credit, calculated as the sum of notes payable, advance receipts, and accounts payable, scaled by total assets. Additionally, we introduce the variable *TTC* as an alternative measurement of trade credit, which includes notes receivable, notes payable, accounts payable, accounts receivable, advance receipts, and advance payments, divided by total assets. Columns (3)-(4) of Table 4 display that CIO is positively associated with trade credit. These findings indicate that firms with CIO strengthen their market power through strategic alliances, enabling them to gain more trade credit. Trade credit intensifies risk interconnectedness among firms, as the default risk of a single firm can propagate to upstream

Table 4
Channel analysis: the risk interconnectedness mechanism.

Variables	(1) Stock price synchronicity	(2)	(3) Trade credit	(4)
	SYN1	*SYN2*	*TC*	*TTC*
LnNumCon	0.197***	0.195***	0.005**	0.007*
	(7.803)	(8.868)	(1.995)	(1.721)
Size	0.178***	0.145***	−0.007***	−0.027***
	(8.128)	(7.972)	(−3.234)	(−6.457)
Beta	0.182***	0.145***	0.000	0.000
	(18.226)	(17.899)	(0.211)	(0.465)
ListAge	0.238***	0.108***	−0.003	0.009**
	(7.837)	(4.247)	(−1.506)	(2.420)
Lev	−0.653***	−0.545***	0.228***	0.316***
	(−9.346)	(−9.228)	(29.373)	(23.699)
TobinQ	−0.097***	−0.080***	0.000	0.001
	(−10.725)	(−10.559)	(0.169)	(0.994)
ROE	−0.046	−0.053	0.048***	0.113***
	(−0.721)	(−0.974)	(8.229)	(11.367)
Dual	0.010	0.002	−0.003	−0.002
	(0.486)	(0.110)	(−1.604)	(−0.865)
Indep	0.092	0.162	−0.005	−0.036
	(0.468)	(0.987)	(−0.356)	(−1.526)
Top10	−0.673***	−0.686***	−0.010	0.005
	(−6.632)	(−7.900)	(−1.022)	(0.282)
Mshare	0.005***	0.003***	−0.000	0.000
	(5.234)	(4.611)	(−0.483)	(0.495)
constant	−3.769***	−2.779***	0.239***	0.788***
	(−8.103)	(−7.152)	(4.826)	(8.665)
IE	YES	YES	YES	YES
YE	YES	YES	YES	YES
Adj-R^2	0.361	0.380	0.188	0.160
N	21,568	21,568	22,515	22,515

Note: This table reports the results of the risk interconnectedness mechanism through which common institutional ownership increases the NFFs' systemic risk in China. *, **, and *** denote the significance levels of 10 %, 5 %, and 1 %, respectively. The values in parentheses represent t-statistics adjusted for robust standard errors. Here, *IE* refers to individual fixed effects, while *YE* denotes time fixed effects.

and downstream firms through the "balance sheet contagion" effect, ultimately raising the probability of systemic risk events. These results further validate the remaining part of Hypothesis 2 by confirming the risk interconnectedness mechanism through which CIO increases NFFs' systemic risk. Consequently, Hypothesis 2 is fully supported.

4.3. Addressing endogeneity and sample selection issues

The impact of CIO on NFFs' systemic risk may be subject to endogeneity concerns. First, the degree of CIO in NFFs may not be random; instead, it could be influenced by factors that also affect systemic risk, leading to selection bias. Second, omitted variables may simultaneously impact both institutional investors' decisions and firm systemic risk. To tackle these concerns, we utilize four methods: instrumental variable approach, PSM analysis, Heckman two-step model, and difference-in-differences analysis.

Our analysis initially assumes that CIO exogenously influences NFFs' systemic risk. However, omitted variables may affect both institutional investment decisions and firm systemic risk, resulting in biased results. To tackle this issue empirically, we employ a two-stage instrumental variable model. We utilize whether a company belongs to the CSI 300 Index (*LnNumCon_IV*) as the instrumental variable (Gao et al., 2019; Liu and He, 2024). Institutional investors typically favor listed firms with large market capitalization, high external ownership, and transparent information disclosure. Since the CSI 300 Index comprises 300 leading Chinese "blue-chip" companies, institutional investors prefer to invest in these companies. Therefore, whether a company belongs to the CSI 300 Index (*LnNumCon_IV*) influences institutional investors' decisions, thereby affecting CIO. This satisfies the relevance condition. Meanwhile, whether a company belongs to the CSI 300 Index is not determined by its systemic risk, fulfilling the exclusion condition.

Columns (1)-(2) of Table 5 display the instrumental variable test findings. The first-stage result confirms that the instrumental variable positively and significantly affects CIO, and the *F*-test rejects the null hypothesis of weak instruments. The second-stage result indicates that CIO significantly and positively impacts firm systemic risk, consistent with our baseline findings. Although no method can completely ensure the validity of the instrument, our instrumental variable analysis suggests that endogeneity is unlikely to bias our baseline findings.

Note that firms with higher CIO may have different characteristics compared to those firms with lower or no CIO. These differences in characteristics may lead to different degrees of firm systemic risk. To solve the endogeneity issue, we re-estimate model (1) using a PSM sample. We match treated firms (with CIO) to control firms (without CIO) using the control variables specified in model (1). We employ one-to-three matching with the nearest neighbor method. The coefficient for *LnNumCon* in Column (3) of Table 5 remains significant and positive, reinforcing our baseline findings.

Table 5
Endogeneity checks: IV, PSM-OLS, and Heckman two-step model.

Variables	(1) Instrumental variable estimation	(2)	(3) PSM-OLS	(4) Heckman two-step model
	LnNumCon	△*CoVaR*	△*CoVaR*	△*CoVaR*
LnNumCon_IV	0.078***			
	(4.28)			
LnNumCon		0.628***	0.041***	0.034**
		(2.65)	(2.881)	(2.479)
Size	0.108***	0.043	0.120***	0.123***
	(13.85)	(1.48)	(5.918)	(4.460)
Beta	0.008***	0.005	0.007	0.006
	(3.04)	(1.25)	(1.079)	(0.782)
ListAge	−0.068***	0.099***	0.072***	0.079*
	(−9.25)	(5.29)	(2.941)	(1.864)
Lev	−0.101***	−0.155***	−0.300***	−0.243***
	(−4.86)	(−3.83)	(−4.206)	(−3.015)
TobinQ	0.039***	−0.036***	−0.010	0.000
	(12.21)	(−3.56)	(−1.593)	(0.047)
ROE	−0.086***	0.145***	0.080	0.114
	(−4.47)	(3.95)	(1.271)	(1.586)
Dual	0.013**	−0.004	0.004	0.007
	(2.08)	(−0.43)	(0.209)	(0.349)
Indep	0.085	−0.142*	−0.084	−0.088
	(1.59)	(−1.66)	(−0.457)	(−0.453)
Top10	0.243***	−0.558***	−0.455***	−0.384***
	(7.42)	(−7.87)	(−4.726)	(−3.152)
Mshare	0.001***	−0.001**	−0.000	−0.002
	(4.46)	(−2.11)	(−0.306)	(−1.218)
imr				−0.070**
				(−2.579)
Constant			−0.476	−0.411
			(−1.257)	(−0.650)
IE	YES	YES	YES	YES
YE	YES	YES	YES	YES
F-test (1st stage)	18.31	–	–	–
LM(p-value)	0.0000	–	–	–
Adj-R^2	–	0.694	0.772	0.782
N	19,296	19,296	5441	3281

Note: This table reports the endogeneity test of IV, PSM-OLS, and Heckman two-step sample selection. *, **, and *** denote the significance levels of 10 %, 5 %, and 1 %, respectively. The values in parentheses represent t-statistics adjusted for robust standard errors. Here, *IE* refers to individual fixed effects, while *YE* denotes time fixed effects.

We use a difference-in-differences (DID) method to further alleviate endogeneity issues. To establish causality, we examine whether a firm's systemic risk significantly rises after it is first held by CIBs within the sample period. To efficiently identify the causal effect, we define the treatment group as firms that are first held by CIBs in 2013 or later and remain continuously held by CIBs for at least three consecutive years thereafter. Sun and Abraham (2021) propose that when treatment timing varies among firms, never-treated firms are the appropriate control group. Therefore, we designate firms without any CIO during the sample period as the control group. The DID model is as follows:

$$\Delta CoVaR_{i,t} = \alpha_0 + \alpha_1 Treat_i \times Post_{t-1} + \alpha_2 Controls_{i,t-1} + \mu_i + \lambda_t + \varepsilon_{i,t} \tag{6}$$

where *Treat* equals 1 for treatment firms and 0 for control firms, and *Post* equals 1 in the year a firm is first held by CIBs and in all subsequent years, and 0 otherwise. The coefficient for *Treat*×*Post* reflects the net impact of CIO on NFFs' systemic risk.

Column (1) of Table 6 displays the difference-in-differences regression findings. The significantly positive coefficient for *Treat*×*Post* supports our hypothesis that CIO increases NFFs' systemic risk. We perform a parallel trend test to confirm the comparability of the treatment and control groups prior to the initial treatment. As a result, we construct interaction terms of the year dummy variable reflecting the years before and after first having CIO with the treatment variable. In Column (2) of Table 6, the coefficients for *Pre_2* and *Pre_3* are statistically insignificant, passing the parallel trends test. Meanwhile, the coefficients for *Post_1* to *Post_3* are significantly positive, supporting our DID results. In addition, the positive but insignificant coefficient for *Cur_0* suggests a lagged impact of CIO on NFFs' systemic risk exhibits a lag. This finding validates the rationale for lagging all explanatory variables by 1 period in the baseline regression.

Sample selection bias is another potential concern for our empirical analysis. Specifically, the observed positive effect of CIO on firm systemic risk may result from unobserved firm characteristics correlated with both firm systemic risk and CIO. We employ the Heckman two-step model to alleviate this concern. We first compute the inverse Mills ratio (*IMR*) to examine whether lagged firm characteristics (i.e., the aforementioned control variables) influence CIO. We employ lagged firm characteristics because institutional

Table 6
Endogeneity check: DID.

Variables	(1) △*CoVaR*	(2) △*CoVaR*
Treat×Post	0.076**	
	(2.420)	
Size	0.106***	0.025***
	(10.959)	(2.608)
Beta	0.011***	0.048***
	(3.127)	(11.656)
ListAge	0.060***	0.110***
	(5.468)	(8.508)
Lev	−0.219***	−0.128***
	(−6.396)	(−3.913)
TobinQ	−0.015***	−0.059***
	(−3.614)	(−12.994)
ROE	0.102***	−0.123***
	(3.057)	(−4.147)
Dual	0.005	−0.006
	(0.465)	(−0.557)
Indep	−0.059	−0.104
	(−0.654)	(−1.077)
Top10	−0.442***	−0.351***
	(−8.786)	(−7.436)
Mshare	−0.000	0.000
	(−0.405)	(0.030)
Pre_3		0.069
		(1.442)
Pre_2		0.046
		(0.935)
Cur_0		0.032
		(0.738)
Post_1		0.106***
		(2.668)
Post_2		0.140***
		(3.356)
Post_3		0.144***
		(3.493)
constant	−0.161	1.846***
	(−0.785)	(9.138)
IE	YES	YES
YE	YES	YES
Adj-R^2	0.734	0.697
N	15,578	18,145

Note: This table reports the endogeneity test of DID, in which column (2) reports the results of the parallel trend test. *, **, and *** denote the significance levels of 10 %, 5 %, and 1 %, respectively. The values in parentheses represent t-statistics adjusted for robust standard errors. Here, *IE* refers to individual fixed effects, while *YE* denotes time fixed effects.

investors make investment decisions based on previously disclosed indicators. According to Lennox et al. (2012), at least one exclusion restriction variable needs to be included in the Heckman first stage. Therefore, we include whether a company belongs to the CSI 300 Index (*LnNumCon_IV*) as the exclusion restriction variable. We then include *IMR* as a control variable in model (1) to test the impact of potential selection bias on the baseline conclusions. Column (4) of Table 5 shows that the coefficient for *IMR* is significant, suggesting the presence of selection bias in CIO. Furthermore, after including the *IMR* as a control variable, α_1 remains significantly positive. Thus, after controlling for selection bias, our conclusion still holds.

4.4. Other robustness checks

4.4.1. Alternative measurements of main variables

In model (1), we employ *ΔCoVaR* as the primary measure to assess NFFs' systemic risk. To examine if systemic risk measurement influences the relationship between CIO and NFFs' systemic risk, we employ Marginal Expected Shortfall (*MES*) as an alternative measurement. *MES* evaluates a firm's expected loss when market returns decline, focusing on measuring systemic risk through the lens of corporate vulnerability. *MES* serves as a valuable complement to *ΔCoVaR* in quantifying systemic risk. In addition, we employ the annual median of *ΔCoVaR* (*ΔCoVaR_P50*) and *MES* (*MES_P50*) for each NFF to measure systemic risk. Considering the limitations of individual indicators, we employ principal component analysis to formulate a new indicator, *PcaSysRisk*, predicated on three metrics: *VaR, ΔCoVaR,* and *MES*. The coefficients for *LnNumCon* remain significantly positive in Table 7, confirming the robustness of our

Table 7
Robustness test: replacing the measurement of explained variables.

Variables	(1) *ΔCoVaR_P50*	(2) *MES*	(3) *MES_P50*	(4) *PcaSysRisk*
LnNumCon	0.042***	0.211***	0.222***	0.078***
	(3.922)	(5.553)	(5.911)	(4.572)
Size	0.116***	0.295***	0.288***	0.126***
	(13.819)	(9.570)	(9.642)	(9.004)
Beta	0.010***	0.102***	0.093***	0.042***
	(3.116)	(8.348)	(7.690)	(7.611)
ListAge	0.051***	−0.269***	−0.260***	−0.097***
	(5.309)	(−6.829)	(−6.809)	(−5.356)
Lev	−0.215***	0.004	−0.033	0.031
	(−7.407)	(0.035)	(−0.287)	(0.584)
TobinQ	−0.008**	0.074***	0.059***	0.041***
	(−2.434)	(5.573)	(4.611)	(7.138)
ROE	0.074**	−0.191*	−0.095	−0.163***
	(2.548)	(−1.653)	(−0.850)	(−3.122)
Dual	0.003	−0.015	−0.024	−0.005
	(0.328)	(−0.451)	(−0.732)	(−0.352)
Indep	−0.056	−0.376	−0.537*	−0.139
	(−0.698)	(−1.284)	(−1.840)	(−1.057)
Top10	−0.399***	−0.594***	−0.585***	−0.217***
	(−8.955)	(−3.684)	(−3.798)	(−2.923)
Mshare	−0.000	−0.000	−0.000	−0.000
	(−0.633)	(−0.001)	(−0.066)	(−0.437)
Constant	−0.470***	0.605	0.768	−2.581***
	(−2.599)	(0.933)	(1.218)	(−8.834)
IE	YES	YES	YES	YES
YE	YES	YES	YES	YES
Adj-R^2	0.694	0.436	0.393	0.637
N	19,305	19,353	19,305	19,353

Note: This table reports the robustness test of replacing the measurement of explained variables. *, **, and *** denote the significance levels of 10 %, 5 %, and 1 %, respectively. The values in parentheses represent t-statistics adjusted for robust standard errors. Here, *IE* refers to individual fixed effects, while *YE* denotes time fixed effects.

conclusions.

Following He and Huang (2017), we construct four alternative measurements for CIO: *LnAvgNum, LnNumCross, TotalCross,* and *CrossDum. LnAvgNum* denotes the number of same-industry firms connected to firm *i* by averaging a CIB. *LnNumCross* signifies the number of firm *i*'s CIBs. Similar to *LnNumCon*, both metrics reflect the extent of firm *i*'s connections to other same-industry firms through CIO. *TotalCross* denotes the total proportion of firm *i*'s shares held by all CIBs, reflecting their potential control strength over the firm. For each measurement, we first calculate quarterly values and then average them over four quarters to derive annual indicators. To mitigate skewness in *LnAvgNum* and *LnNumCross*, we apply a logarithmic transformation. Finally, *CrossDum* represents a binary variable set to 1 if at least one CIB holds firm *i* within a year; otherwise, it equals 0. The coefficients for the variables of interest in columns (1)-(4) of Table 8 are significantly positive across all CIO measurements, reinforcing the robustness of the main conclusions.

4.4.2. Alternative institutional blockholder thresholds

In the preceding section, we apply a 5 % ownership threshold to identify institutional blockholders when calculating CIO metrics. In line with Gao et al. (2019), we also apply a 3 % threshold to include additional institutional shareholders with smaller shareholdings. Additionally, following Wang et al. (2023), we apply a 10 % threshold to identify institutional blockholders with substantial ownership, thereby reducing noise in CIO measurement.

As professional participants in capital markets, CIBs possess informational advantages. These advantages allow them to influence corporate management decisions even with small ownership stakes. Therefore, we expect that CIO identified at 3 % and 10 % also significantly impact NFFs' systemic risk, with larger ownership stakes magnifying this effect. The coefficients for the main explanatory variable in columns (5)-(6) of Table 8 are significantly positive, reinforcing the robustness of our baseline results. Furthermore, as the threshold increases, the coefficient also rises, suggesting that larger ownership stakes enhance investors' influence on corporate decision-making.

4.4.3. Controlling for industry and regional differences

To capture any within-industry-year and within-area-year variation regarding CIO that may influence NFFs' systemic risk, we re-estimate model (1) after incorporating industry×year and province×year interaction fixed effects while controlling for individual fixed effects. Table 9 indicates that the coefficients for *LnNumCon* remain significantly positive after accounting for industry and regional differences, further confirming the robustness of our conclusions.

Table 8
Robustness test: replacing the measurement of explanatory variables and changing the thresholds of ownership proportions.

Variables	(1) △*CoVaR*	(2) △*CoVaR*	(3) △*CoVaR*	(4) △*CoVaR*	(5) △*CoVaR*	(6) △*CoVaR*
TotalCross	0.170**					
	(2.318)					
LnAvgNum		0.045***				
		(3.942)				
LnNumCross			0.073***			
			(2.991)			
CrossDum				0.034**		
				(2.459)		
LnNumCon1					0.018***	
					(2.700)	
LnNumCon2						0.084***
						(3.241)
Size	0.116***	0.112***	0.114***	0.114***	0.113***	0.115***
	(13.550)	(12.972)	(13.205)	(13.327)	(12.947)	(13.489)
Beta	0.010***	0.010***	0.010***	0.010***	0.010***	0.010***
	(3.235)	(3.148)	(3.206)	(3.225)	(3.187)	(3.234)
ListAge	0.058***	0.060***	0.059***	0.059***	0.060***	0.058***
	(5.889)	(6.080)	(5.994)	(5.962)	(6.019)	(5.885)
Lev	−0.225***	−0.221***	−0.223***	−0.223***	−0.221***	−0.224***
	(−7.414)	(−7.282)	(−7.346)	(−7.349)	(−7.296)	(−7.360)
TobinQ	−0.011***	−0.012***	−0.011***	−0.011***	−0.012***	−0.011***
	(−3.039)	(−3.431)	(−3.228)	(−3.165)	(−3.354)	(−3.069)
ROE	0.094***	0.096***	0.096***	0.095***	0.094***	0.096***
	(3.163)	(3.225)	(3.224)	(3.194)	(3.152)	(3.224)
Dual	0.004	0.004	0.004	0.004	0.003	0.004
	(0.437)	(0.385)	(0.410)	(0.406)	(0.379)	(0.459)
Indep	−0.091	−0.089	−0.090	−0.088	−0.089	−0.092
	(−1.107)	(−1.080)	(−1.097)	(−1.074)	(−1.085)	(−1.126)
Top10	−0.416***	−0.421***	−0.419***	−0.417***	−0.417***	−0.419***
	(−9.175)	(−9.268)	(−9.240)	(−9.206)	(−9.196)	(−9.232)
Mshare	−0.000	−0.000	−0.000	−0.000	−0.000	−0.000
	(−0.932)	(−0.998)	(−0.956)	(−0.935)	(−0.969)	(−0.951)
constant	−0.374**	−0.291	−0.331*	−0.344*	−0.310*	−0.357**
	(−2.066)	(−1.587)	(−1.806)	(−1.892)	(−1.675)	(−1.971)
IE	YES	YES	YES	YES	YES	YES
YE	YES	YES	YES	YES	YES	YES
Adj-R^2	0.740	0.741	0.741	0.740	0.740	0.741
N	19,353	19,353	19,353	19,353	19,353	19,353

Note: This table reports the robustness test of replacing the measurement of explanatory variables and changing the thresholds of ownership proportions. *, **, and *** denote the significance levels of 10 %, 5 %, and 1 %, respectively. The values in parentheses represent t-statistics adjusted for robust standard errors. Here, *IE* refers to individual fixed effects, while *YE* denotes time fixed effects.

4.4.4. Controlling for other shocks

First, the 2015 stock market crash caused extreme volatility in China's financial markets and significant adverse shocks to NFFs, ultimately increasing their systemic risk. To mitigate the influence of the 2015 stock market crash, we modify model (1) by incorporating an interaction term between Ln*NumCon* and a crisis dummy variable (*Crisis*), where *Crisis* equals 1 for 2015 and 0 otherwise. Column (1) of Table 10 displays that the coefficient for *LnNumCon* remains significantly positive after controlling for the stock market crash, further validating the robustness of our main conclusions.

Second, supply-side structural reforms, focusing on capacity reduction, inventory reduction, deleveraging, cost reduction, and remedying deficiencies, aim to restructure the economy and mitigate the buildup of corporate vulnerabilities, potentially affecting NFFs' systemic risk. To isolate this effect, we exclude samples from manufacturing sectors significantly affected by the reform (e.g., coal and steel industries) and re-estimate the model (1). Column (2) of Table 10 confirms that our baseline findings remain robust after considering the influence of supply-side structural reforms.

Third, monetary policy fluctuations affect firms' external financing conditions, thereby influencing their systemic risk. A tightening monetary policy typically raises financing costs and restricts access to external capital, thereby heightening firms' systemic risk exposure. Conversely, an accommodative monetary policy can alleviate financing pressures and subsequently mitigate systemic risk. Consequently, we incorporate the monetary policy uncertainty index (*MPU*) introduced by Huang and Luk (2020) into model (1). Column (3) of Table 10 demonstrates that our baseline results remain robust even after accounting for monetary policy fluctuations, further substantiating our primary conclusions' validity.

Fourth, local government debt governance may mitigate corporate credit mismatches by lowering financing costs and enhancing credit availability (Luo and Li, 2025), thereby reducing NFFs' systemic risk. We incorporate a time dummy variable, *Post15*, indicating the formal implementation of the 2015 New Budget Law, and use the average ratio of local government investment bond issuance to fiscal revenue (*Citybond*) from 2010 to 2014 as a treatment variable. Their interaction term, *Citybond_Post15*, serves as a proxy for local

Table 9
Robustness test: controlling for industry and regional differences.

Variables	(1) △*CoVaR*	(2) △*CoVaR*	(3) △*CoVaR*
LnNumCon	0.038***	0.038***	0.034***
	(3.282)	(3.525)	(2.961)
Size	0.097***	0.100***	0.088***
	(10.104)	(11.647)	(9.269)
Beta	0.010***	0.009***	0.010***
	(2.812)	(2.919)	(2.783)
ListAge	0.073***	0.067***	0.076***
	(6.448)	(6.828)	(6.690)
Lev	−0.174***	−0.206***	−0.173***
	(−5.125)	(−7.015)	(−5.210)
TobinQ	−0.013***	−0.017***	−0.017***
	(−3.463)	(−4.918)	(−4.727)
ROE	0.113***	0.075***	0.096***
	(3.349)	(2.580)	(2.871)
Dual	0.003	0.004	0.001
	(0.356)	(0.398)	(0.110)
Indep	−0.079	−0.092	−0.060
	(−0.886)	(−1.114)	(−0.675)
Top10	−0.395***	−0.383***	−0.382***
	(−8.038)	(−8.402)	(−7.670)
Mshare	−0.001*	−0.000	−0.001**
	(−1.914)	(−1.032)	(−2.019)
Constant	−0.183	−0.214	0.008
	(−0.890)	(−1.156)	(0.038)
IE	YES	YES	YES
YE	NO	NO	NO
CE×*YE*	NO	YES	YES
AE×*YE*	YES	NO	YES
Adj-R^2	0.862	0.851	0.873
N	17,864	19,289	17,857

Note: This table reports the robustness test of controlling for industry and regional differences. *, **, and *** denote the significance levels of 10 %, 5 %, and 1 %, respectively. The values in parentheses represent t-statistics adjusted for robust standard errors. *IE, YE, AE*, and *CE* denote individual, year, province, and industry fixed effects, respectively.

government debt governance. We then incorporate this variable into model (1). Column (4) of Table 10 indicates that our primary findings remain robust after accounting for the influence of local government debt governance.

Finally, capital market liberalization enables eligible foreign investors to freely access Chinese capital markets, thereby deepening the integration of domestic and global capital markets. This enhanced connectivity facilitates the transmission of external shocks across markets, ultimately intensifying NFFs' systemic risk (Li et al., 2023a). We introduce a dummy variable, $Hsgt_{i,t}$, into model (1), which indicates whether firm *i* is included in the Shanghai-Hong Kong or Shenzhen-Hong Kong Stock Connect program in year *t*. It equals 1 if the firm is included that year and 0 otherwise. Column (5) of Table 10 further corroborates the baseline findings after accounting for the influence of capital market liberalization.

5. Further analyses

5.1. Heterogeneity analysis

5.1.1. The role of stock liquidity

During periods of significant market volatility, stocks with lower liquidity are harder to sell quickly due to higher transaction costs, making them more vulnerable to price shocks. Consequently, firms with lower stock liquidity tend to exhibit higher systemic risk (Nadarajah et al., 2021). Our analysis suggests that CIO enhances risk interconnectedness among firms by increasing stock price synchronicity and trade credit. Therefore, when NFFs with low stock liquidity face market shocks, CIO amplifies its individual risk, leading to higher systemic risk. Thus, the influence of CIO on NFFs' systemic risk is more significant in NFFs with lower stock liquidity. Following Hsu et al. (2020), we employ the tradable turnover ratio to characterize stock liquidity. We use the median of this indicator to classify the sample into high and low stock liquidity groups. We then conduct separate regression analyses for each group. Columns (1)-(2) of Table 11 suggest that in NFFs with lower stock liquidity, the influence of CIO on NFFs' systemic risk is more significant. Moreover, the coefficient difference is statistically significant, as demonstrated by the bootstrap test.

5.1.2. The role of transparency

CIBs have the motivation to collude through leverage manipulation and earnings management, thereby concealing firms' true financial condition and exacerbating information asymmetry between firms and external investors. In firms with lower transparency,

Table 10
Robustness test: controlling for other shocks.

Variables	(1) △*CoVaR*	(2) △*CoVaR*	(3) △*CoVaR*	(4) △*CoVaR*	(5) △*CoVaR*
LnNumCon	0.040***	0.038***	0.041***	0.038***	0.034***
	(3.644)	(3.525)	(3.737)	(3.400)	(3.069)
LnNumCon_Crisis1	0.059				
	(1.398)				
Crisis1	0.349***				
	(23.274)				
MPU			0.005***		
			(32.319)		
Citybond_Post15				−0.208**	
				(−1.963)	
Hsgt					0.062***
					(6.627)
Size	0.112***	0.111***	0.112***	0.116***	0.094***
	(13.013)	(12.697)	(13.005)	(13.051)	(10.541)
Beta	0.010***	0.011***	0.010***	0.009***	0.011***
	(3.192)	(3.426)	(3.157)	(2.861)	(3.387)
ListAge	0.060***	0.060***	0.060***	0.053***	0.064***
	(6.039)	(6.025)	(6.072)	(5.157)	(6.460)
Lev	−0.221***	−0.212***	−0.221***	−0.225***	−0.202***
	(−7.290)	(−6.924)	(−7.285)	(−7.185)	(−6.665)
TobinQ	−0.012***	−0.013***	−0.012***	−0.012***	−0.014***
	(−3.335)	(−3.768)	(−3.399)	(−3.283)	(−4.084)
ROE	0.097***	0.083***	0.096***	0.086***	0.092***
	(3.252)	(2.729)	(3.222)	(2.802)	(3.112)
Dual	0.004	0.004	0.004	0.006	0.004
	(0.391)	(0.439)	(0.385)	(0.628)	(0.417)
Indep	−0.089	−0.118	−0.089	−0.050	−0.087
	(−1.089)	(−1.413)	(−1.085)	(−0.606)	(−1.066)
Top10	−0.420***	−0.402***	−0.420***	−0.423***	−0.406***
	(−9.264)	(−8.780)	(−9.258)	(−9.092)	(−8.982)
Mshare	−0.000	−0.000	−0.000	−0.000	−0.000
	(−0.986)	(−0.952)	(−0.993)	(−1.195)	(−1.306)
Constant	−0.298	−0.266	−1.307***	−0.368*	0.097
	(−1.627)	(−1.436)	(−6.662)	(−1.950)	(0.513)
IE	YES	YES	YES	YES	YES
YE	YES	YES	YES	YES	YES
Adj-R^2	0.741	0.743	0.741	0.744	0.742
N	19,353	18,900	19,353	18,184	19,316

Note: This table reports the robustness test of controlling for other shocks. *, **, and *** denote the significance levels of 10 %, 5 %, and 1 %, respectively. The values in parentheses represent t-statistics adjusted for robust standard errors. Here, *IE* refers to individual fixed effects, while *YE* denotes time fixed effects.

such manipulative practices are harder to detect, making it easier for CIBs to collude and exacerbating systemic risk accumulation. Moreover, minority shareholders and creditors may struggle to accurately assess the true financial and operational conditions of such opaque firms, thereby hindering effective external oversight and further accelerating systemic risk accumulation. Consequently, lower transparency amplifies the impact of CIO on NFFs' systemic risk.

To examine how transparency influences the association between CIO and NFFs' systemic risk, we employ the research report attention to characterize the transparency. We gauge it by counting the number of research reports analyzing the company during a specific year. We use the median of this indicator to classify the sample into high and low transparency groups. We then conduct regression analyses separately for each group. Columns (3)-(4) of Table 11 display the findings. The empirical p-value suggests a significant difference between the two sub-samples, showing that CIO has a stronger effect on firm systemic risk in low-transparency NFFs. The finding underscores the critical role of transparency in curbing manipulative behavior and strengthening external oversight. In high-transparency firms, accurate and timely information deters collusion and enhances stakeholder oversight, thereby mitigating systemic risk. Conversely, low-transparency environments provide cover for manipulative practices, which weakens external scrutiny and magnifies the effect of CIO on systemic risk.

5.1.3. The role of product market competition

Industries with low product market competition typically feature a concentrated market structure, where a few dominant firms hold significant market power. This concentration heightens market dependence on specific firms, making the entire industry more vulnerable to adverse economic or market conditions. Furthermore, insufficient competition may hinder firm investment and innovation, diminishing their ability to adapt to economic fluctuations (Gu, 2016). The interconnections fostered by CIO are particularly pronounced in industries with lower product market competition. Therefore, strong firm linkages in these industries foster collusive behavior, enabling firms to extract excess profits through manipulation, thereby increasing corporate fragility. Moreover, strong

Table 11
Heterogeneity analysis: stock liquidity and transparency.

Variables	(1) High stock liquidity	(2) Low stock liquidity	(3) High transparency	(4) Low transparency
LnNumCon	0.022	0.057***	0.034***	0.069*
	(1.244)	(4.099)	(2.750)	(1.786)
Size	0.085***	0.144***	0.150***	0.061***
	(7.602)	(9.928)	(10.500)	(3.953)
Beta	0.001	0.024***	0.010**	0.009*
	(0.335)	(3.818)	(2.027)	(1.836)
ListAge	0.060***	0.036	0.043***	0.094***
	(4.657)	(1.494)	(2.977)	(5.294)
Lev	−0.180***	−0.278***	−0.277***	−0.212***
	(−4.382)	(−5.511)	(−6.004)	(−4.316)
TobinQ	−0.006	−0.016***	−0.013***	−0.015**
	(−1.273)	(−2.736)	(−2.736)	(−2.291)
ROE	0.068	0.110**	0.063	0.131***
	(1.592)	(2.326)	(1.223)	(3.179)
Dual	−0.002	0.017	−0.002	0.014
	(−0.214)	(1.133)	(−0.172)	(0.901)
Indep	−0.118	−0.085	−0.134	−0.093
	(−1.072)	(−0.683)	(−0.989)	(−0.789)
Top10	−0.413***	−0.382***	−0.350***	−0.441***
	(−6.578)	(−5.403)	(−4.948)	(−6.284)
Mshare	0.000	−0.001	−0.001	−0.000
	(0.681)	(−1.518)	(−1.230)	(−0.393)
constant	0.231	−0.979***	−1.084***	0.754**
	(0.977)	(−3.125)	(−3.497)	(2.339)
IE	YES	YES	YES	YES
YE	YES	YES	YES	YES
Empirical p-value	0.000***		0.000***	
Adj-R^2	0.756	0.747	0.757	0.730
N	9836	9517	10,106	9247

Note: This table reports the results of the heterogeneity analysis. *, **, and *** denote the significance levels of 10 %, 5 %, and 1 %, respectively. The values in parentheses represent t-statistics adjusted for robust standard errors. Here, *IE* refers to individual fixed effects, while *YE* denotes time fixed effects. The empirical p-value is obtained from a bootstrap test on the coefficient differences.

inter-firm connections intensify risk interconnectedness, amplifying shock propagation and exacerbating systemic risk.

To examine how product market competition influences the association between CIO and NFFs' systemic risk, we employ the Herfindahl-Hirschman Index (*HHI*) as a metric for assessing product market competition, defined as the sum of the squared market shares of all same-industry companies. A higher *HHI* indicates increased industry concentration and reduced product market competition. We utilize this indicator's median to categorize the sample into two groups. We then perform regression analyses separately for each group. Columns (1)-(2) of Table 12 display that in firms characterized by lower product market competition, the influence of CIO on NFFs' systemic risk is more significant. This indicates that in these industries, inter-firm risk interconnectedness is greater, and CIBs are more prone to collusive behavior, thereby increasing systemic risk. Furthermore, the coefficient difference is statistically significant, as confirmed by the bootstrap test.

5.1.4. The role of financing constraints

Firms experiencing higher financing constraints often struggle to secure funds through traditional banking channels. In such cases, CIBs may have incentives to manipulate firms to conceal their true financial status through leverage manipulation, earnings management, and other means to obtain external financing. These behaviors increase firms' financial fragility. Moreover, firms facing higher financing constraints are more vulnerable to economic shocks due to their limited financial flexibility. Since CIO strengthens inter-firm risk interconnectedness, external shocks affecting financially constrained firms are more likely to propagate to other firms, thereby amplifying systemic risk. We utilize the WW index to assess NFFs' financing constraints, as proposed by Whited and Wu (2006). By categorizing the sample according to the annual median value, we examine how the influence of CIO on corporate systemic risk differs between NFFs with varying financing constraints. Columns (3)-(4) of Table 12 suggest that for NFFs with higher financing constraints, CIO exerts a stronger impact on their systemic risk. These results show that under financing constraints, CIBs have stronger incentives to manipulate firms and conceal their true financial conditions, thereby increasing their risk exposure. Moreover, the difference is statistically significant, as evidenced by the bootstrap coefficient difference test.

5.2. An examination of tail risk spillover of CIO peer firms

To further explore the impact of CIO on NFFs' systemic risk, we refine our study by focusing exclusively on firms with CIO. Following Van Oordt and Zhou (2019), we decompose the systemic risk indicator into systemic linkage and individual tail risk. Since CIO inherently reflects a form of linkage, we concentrate solely on whether the tail risk of CIO peer firms spills over to the focal firm

Table 12
Heterogeneity analysis: product market competition and financing constraints.

Variables	(1) High market competition	(2) Low market competition	(3) High financing constraints	(4) Low financing constraints
LnNumCon	0.017	0.055***	0.072***	0.036***
	(1.203)	(3.120)	(3.237)	(2.697)
Size	0.119***	0.091***	0.094***	0.188***
	(8.897)	(8.145)	(7.149)	(11.195)
Beta	0.005	0.014***	0.011**	0.014***
	(0.857)	(3.134)	(2.551)	(2.609)
ListAge	0.080***	0.052***	0.066***	0.048**
	(4.707)	(3.994)	(5.050)	(2.539)
Lev	−0.209***	−0.236***	−0.194***	−0.267***
	(−4.493)	(−5.469)	(−4.925)	(−4.686)
TobinQ	−0.017***	−0.009*	−0.012***	−0.019***
	(−3.292)	(−1.806)	(−2.659)	(−2.842)
ROE	0.088*	0.112***	0.140***	0.104*
	(1.799)	(2.649)	(3.758)	(1.861)
Dual	0.025*	−0.019	−0.007	0.009
	(1.770)	(−1.591)	(−0.599)	(0.553)
Indep	−0.111	−0.132	0.031	−0.092
	(−0.986)	(−1.004)	(0.314)	(−0.658)
Top10	−0.415***	−0.356***	−0.393***	−0.501***
	(−5.952)	(−5.506)	(−6.214)	(−7.017)
Mshare	−0.001**	0.000	−0.000	0.000
	(−2.396)	(0.344)	(−0.742)	(0.151)
constant	−0.454	0.131	−0.001	−1.882***
	(−1.573)	(0.545)	(−0.004)	(−5.175)
IE	YES	YES	YES	YES
YE	YES	YES	YES	YES
Empirical p-value	0.000***		0.000***	
Adj-R^2	0.752	0.741	0.704	0.789
N	8921	10,432	11,143	8210

Note: This table reports the results of the heterogeneity analysis. *, **, and *** denote the significance levels of 10 %, 5 %, and 1 %, respectively. The values in parentheses represent t-statistics adjusted for robust standard errors. Here, *IE* refers to individual fixed effects, while *YE* denotes time fixed effects. The empirical p-value is obtained from a bootstrap test on the coefficient differences.

within this sample.

We employ two approaches to calculate peer tail risk with the following specifications:

$$Peer\Delta VaR_{i,t} = \frac{\sum_{j=1}^{n} w_{i,j,t}\Delta VaR_{j,t}}{\sum_{j=1}^{n} w_{i,j,t}} \tag{7}$$

$$w_{i,j,t} = \frac{\sum_{f\in F}\left(S_{i,t}^{f}P_{i,t} + S_{j,t}^{f}P_{j,t}\right)}{S_{i,t}P_{i,t} + S_{j,t}P_{j,t}} \tag{8}$$

where n denotes the number of CIO peer firms for firm i. Following Cheng et al. (2023), we utilize the model (8) to calculate the weight w, where F denotes the set of CIBs for firm i. $S_{i,t}^{f}$ represents the shares of firm i owned by CIB f, while $S_{i,t}$ represents firm i's total outstanding shares. $P_{i,t}$ represents firm i's stock price. Additionally, following Foroughi et al. (2021), we adopt an alternative measurement using equal weights. This measure is also calculated using model (7), but w is set to 1.

We conduct our empirical analysis according to the following model:

$\Delta VaR_{i,t} = a_0 + a_1 Peer\Delta VaR_{i,t} + a_2 Controls_{i,t} + \mu_i + \lambda_t + \varepsilon_{i,t}$ (9)

Table 13 shows that firm i's tail risk is positively correlated with its CIO peers' tail risk, confirming that CIO networks propagate individual firms' tail risk and thereby form a systemic risk danger.

6. Conclusions

In this study, we investigated how CIO influences NFFs' systemic risk. We found that CIO increases NFFs' systemic risk. We identify two channels through which CIO positively affects NFFs' systemic risk: exacerbating risk accumulation through firm manipulation and increasing risk interconnectedness among firms. Our baseline findings continue to hold after conducting several robustness tests. Heterogeneity analyses reveal that the relationship between CIO and systemic risk is more significant in NFFs with lower stock liquidity, lower transparency, lower product market competition, and higher financing constraints. Finally, we find that CIO networks propagate individual firms' tail risk and thereby form a systemic risk danger.

Our paper offers several policy implications. First, to mitigate firms' excessive interconnectedness through CIO, regulators should

Table 13
Further analysis: CIO tail risk spillover.

Variables	(1) *ΔVaR*	(2) *ΔVaR*
PeerΔVaR1	0.105***	
	(3.690)	
PeerΔVaR2		0.103***
		(3.779)
Size	−0.050	−0.052
	(−0.358)	(−0.374)
Beta	0.319***	0.319***
	(6.898)	(6.890)
ListAge	−1.506***	−1.499***
	(−4.948)	(−4.927)
Lev	1.567***	1.576***
	(3.230)	(3.259)
TobinQ	0.346***	0.346***
	(6.215)	(6.218)
ROE	0.263	0.264
	(0.606)	(0.607)
Dual	−0.143	−0.143
	(−0.915)	(−0.915)
Indep	1.541	1.503
	(1.245)	(1.223)
Top10	−0.390	−0.396
	(−0.563)	(−0.572)
Mshare	−0.002	−0.002
	(−0.258)	(−0.263)
constant	10.663***	10.730***
	(3.645)	(3.691)
IE	YES	YES
YE	YES	YES
Adj-R^2	0.442	0.442
N	2162	2167

Note: This table reports the results of the analysis on CIO peer tail risk spillover. *, **, and *** denote the significance levels of 10 %, 5 %, and 1 %, respectively. The values in parentheses represent t-statistics adjusted for robust standard errors. Here, *IE* refers to individual fixed effects, while *YE* denotes time fixed effects.

enhance dynamic monitoring and early-warning systems, implement real-time oversight of institutional investors' cross-holdings, and impose caps on CIO holdings in critical industries. Meanwhile, regulators should guide institutional investors to refine their investment strategies and diversify portfolios to reduce dependency on any single industry or enterprise. Second, regulators should closely monitor institutional investors' influence on corporate disclosure, especially in earnings management and leverage manipulation, which may obscure firms' true financial conditions. Policymakers should implement stricter disclosure requirements, mandate regular publication of key financial data, and involve independent auditors to ensure accuracy and transparency. Third, considering the increasing systemic importance of NFFs, regulators should enhance the assessment and supervision of systemic risks in NFFs and develop a comprehensive risk monitoring system. Moreover, within a prudential regulatory framework, regulators should prioritize systemically important firms to achieve targeted supervision and effective risk mitigation.

Although our research deeply explores how CIO affects NFFs' systemic risk, two limitations merit further investigation. First, this paper centers on listed NFFs' systemic risk while overlooking relatively more vulnerable unlisted firms. Future research should seek to quantify the systemic risk of unlisted firms and extend the sample to encompass all real-economy firms. This broader scope would enhance the generalizability of the findings and provide more comprehensive insights for mitigating systemic risk across the real economy. Second, our study focuses on horizontal common ownership and does not explore how systemic risk propagates through vertical ownership or production networks. Future research could examine how vertical ownership structures and production linkages influence the propagation of systemic risk among interconnected firms, providing a more comprehensive understanding of risk contagion mechanisms within affiliated firms.

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CRediT authorship contribution statement

Xiao-Lin Li: Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition,

Formal analysis, Data curation. **Jiawei Hao:** Writing – original draft, Software, Investigation, Data curation. **Liang Liu:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A Variable Definitions

Dependent variables: NFFs' systemic risk measures.	
ΔCoVaR (%)	Delta conditional value-at-risk proposed by Adrian and Brunnermeier (2016).
MES (%)	Marginal Expected Shortfall proposed by Acharya et al. (2017).
PcaSysRisk (%)	A composite index calculated using PCA predicated on three metrics: *VaR, ΔCoVaR*, and *MES*.
Independent variables: Common institutional ownership measures of the NFFs.	
LnNumCon	The natural logarithm of 1 plus the number of same-industry firms linked to the focal firm via its shared institutional blockholder.
LnAvgNum	The natural logarithm of 1 plus the number of same-industry firms connected to the focal firm by averaging a common institutional blockholder.
TotalCross (%)	The aggregate percentage of holdings by all common institutional blockholders in the focal firm.
LnNumCross	The natural logarithm of 1 plus the number of common institutional blockholders.
CrossDum	A binary variable set to 1 if the firm is held by at least one common institutional blockholder within a year; otherwise, it equals 0.
Control variables	
Size	The natural logarithm of total assets.
Beta	The risk assessment coefficient derived from regressing firm stock returns on market stock returns.
ListAge	The natural logarithm of the listing age.
Lev (%)	Total liabilities / total assets ×100 %.
TobinQ (%)	Firm market value / total assets ×100 %.
ROE (%)	Net profit / shareholders' equity ×100 %.
Dual	A binary variable indicating whether the CEO also serves as the Chairman.
Indep (%)	Percentage of independent directors to total number of directors.
Top10 (%)	The proportion of total shares held by firm's top 10 shareholders.
Mshare (%)	The proportion of the company's shares owned by its executives.
Mechanism variables	
LEVM	A leverage manipulation measure computed by the basic XLT-LEVM approach.
ExpLEVM	Another leverage manipulation measure computed by the extended XLT-LEVM approach.
REM	Real earnings management measure proposed by Roychowdhury (2006).
SYN1	Stock price synchronicity measure following An and Zhang (2013).
SYN2	Stock price synchronicity measure following Boubaker et al. (2014).
TC	The sum of notes payable, advance receipts, and accounts payable, scaled by total assets
TTC	The sum of notes payable, advance receipts, accounts payable, notes receivable, accounts receivable, and advance payments, divided by total assets.
Other variables	
PeerΔVaR1	The value-weighted average of CIO peers' tail risk based on *ΔVaR* following Cheng et al. (2023).
PeerΔVaR2	The equal-weighted average of CIO peers' tail risk based on *ΔVaR* following Foroughi et al. (2021).

References

Accornero, M., Cascarino, G., Felici, R., Parlapiano, F., Sorrentino, A.M., 2018. Credit risk in banks' exposures to non-financial firms. Eur. Financ. Manag. 24, 775–791.

Acharya, V.V., Pedersen, L.H., Philippon, T., Richardson, M., 2017. Measuring systemic risk. Rev. Financ. Stud. 30, 2–47.

Adrian, T., Brunnermeier, M.K., 2016. CoVaR. Am. Econ. Rev. 106, 1705.

An, H., Zhang, T., 2013. Stock price synchronicity, crash risk, and institutional investors. J. Corp. Financ. 21, 1–15.

Anginer, D., Demirguc-Kunt, A., Huizinga, H., Ma, K., 2018. Corporate governance of banks and financial stability. J. Financ. Econ. 130, 327–346.

Azar, J., Schmalz, M.C., Tecu, I., 2018. Anticompetitive effects of common ownership. J. Finance 73, 1513–1565.

Benoit, S., Colliard, J.-E., Hurlin, C., Pérignon, C., 2017. Where the risks lie: a survey on systemic risk. Rev. Financ. 21, 109–152.

Berger, A.N., Roman, R.A., Sedunov, J., 2020. Did TARP reduce or increase systemic risk? The effects of government aid on financial system stability. J. Financ. Intermed. 43, 100810.

Boubaker, S., Mansali, H., Rjiba, H., 2014. Large controlling shareholders and stock price synchronicity. J. Bank. Financ. 40, 80–96.

Chen, S., Ma, H., Wu, Q., Zhang, H., 2024. Common institutional ownership and stock price crash risk. Contemp. Account. Res. 41, 679–711.

Chen, Y., Li, Q., Ng, J., Wang, C., 2021. Corporate financing of investment opportunities in a world of institutional cross-ownership. J. Corp. Financ. 69, 102041.

Cheng, C.A., Xie, J., Zhong, Y., 2023. Common institutional blockholders and tail risk. J. Bank. Financ. 148, 106723.

Cheng, X., (Helen) Wang, H., Wang, X., 2022. Common institutional ownership and corporate social responsibility. J. Bank. Financ. 136, 106218.

De George, E.T., Reiter, N., Synn, C., Williams, C.D., 2017. Institutional ownership and the propagation of systemic risk among banks. SSRN Electron. J.

DeAngelo, H., Gonçalves, A.S., Stulz, R.M., 2018. Corporate deleveraging and financial flexibility. Rev. Financ. Stud. 31, 3122–3174.

DesJardine, M.R., Grewal, J., Viswanathan, K., 2023. A rising tide lifts all boats: the effects of common ownership on corporate social responsibility. Organ. Sci. 34, 1716–1735.

Dungey, M., 2022. Non-financial corporations and systemic risk. J. Corp. Financ.

Edmans, A., Levit, D., Reilly, D., 2019. Governance under common ownership. Rev. Financ. Stud. 32, 2673–2719.

Foroughi, P., Marcus, A.J., Nguyen, V., Tehranian, H., 2021. Peer effects in corporate governance practices: evidence from universal demand laws. Rev. Financ. Stud. 35, 132–167.
Gao, G.P., Moulton, P.C., Ng, D.T., 2017. Institutional ownership and return predictability across economically unrelated stocks. J. Financ. Intermed. 31, 45–63.
Gao, K., Shen, H., Gao, X., Chan, K.C., 2019. The power of sharing: evidence from institutional investor cross-ownership and corporate innovation. Int. Rev. Econ. Financ. 63, 284–296.
Gu, L., 2016. Product market competition, R&D investment, and stock returns. J. Financ. Econ. 119, 441–455.
Haghighat, A., Farhangzadeh, B., Haghighat, M., 2015. The impact of institutional ownership on stock price synchronicity and crash risk. Int. J. Bus. Soc. Sci. 6.
He, J., (Jack), Huang, J., 2017. Product market competition in a world of cross-ownership: evidence from institutional blockholdings. Rev. Financ. Stud. 30, 2674–2718.
Hsu, C., Ma, Z., Wu, L., Zhou, K., 2020. The effect of stock liquidity on corporate risk-taking. J. Account. Audit. Finance 35, 748–776.
Huang, Y., Luk, P., 2020. Measuring economic policy uncertainty in China. China Econ. Rev. 59, 101367.
Jin, L., Myers, S.C., 2006. R2 around the world: new theory and new tests. J. Financ. Econ.
Kang, J.-K., Luo, J., Na, H.S., 2018. Are institutional investors with multiple blockholdings effective monitors? J. Financ. Econ. 128, 576–602.
Kiyotaki, N., Moore, J., 2002. Balance-sheet contagion. Am. Econ. Rev. 92, 46–50.
Lennox, C.S., Francis, J.R., Wang, Z., 2012. Selection models in accounting research. Account. Rev. 87, 589–616.
Lewellen, K., Lowry, M., 2021. Does common ownership really increase firm coordination? J. financ. econ. 141, 322–344.
Li, L., Francis, B.B., Hasan, I., 2011. Firms' Real earnings management and subsequent stock price crash risk. SSRN Electron. J.
Li, X.-L., Li, H., Ge, X., Si, D.-K., 2023a. Capital market liberalization and systemic risk of non-financial firms: evidence from Chinese Stock Connect scheme. Pac-Basin. Financ. J. 82, 102190.
Li, X.-L., Wang, L., Kong, D., 2023b. Macro-prudential policy and systemic risk of real estate firms: evidence from China. Financ. Res. Lett. 58, 104518.
Liu, C., He, J., 2024. Does common institutional ownership improve the similarity of behavior between firms? Evidence from China. Pac-Basin. Financ. J. 83, 102225.
Liu, H., Hou, C., 2023. Institutional cross-ownership and trade credit: evidence from China. Corp. Gov. Int. Rev. 31, 845–868.
Luo, J., Li, J., 2025. How does local government debt governance affect corporate credit mismatches? Evidence from implementing the New Budget Law. Financ. Res. Lett. 73, 106576.
Maffett, M., 2012. Financial reporting opacity and informed trading by international institutional investors. J. Account. Econ. 54, 201–220.
Matousek, R., Panopoulou, E., Papachristopoulou, A., 2020. Policy uncertainty and the capital shortfall of global financial firms. J. Corp. Financ. 62, 101558.
Nadarajah, S., Duong, H.N., Ali, S., Liu, B., Huang, A., 2021. Stock liquidity and default risk around the world. J. Financ. Mark. 55, 100597.
Ni, X., Yin, D., 2023. Is institutional common ownership commonly priced? Insights from the cost of equity capital. J. Bank. Financ. 155, 106990.
Park, J., Sani, J., Shroff, N., White, H., 2019. Disclosure incentives when competing firms have common ownership. J. Account. Econ. 67, 387–415.
Ramalingegowda, S., Utke, S., Yu, Y., 2021. Common institutional ownership and earnings management. Contemp. Account. Res. 38, 208–241.
Roychowdhury, S., 2006. Earnings management through real activities manipulation. J. Account. Econ. 42, 335–370.
Shim, J., 2019. Loan portfolio diversification, market structure and bank stability. J. Bank. Financ. 104, 103–115.
Si, D.-K., Li, X.-L., 2022. Shadow banking business and firm risk-taking: evidence from China. Res. Int. Bus. Finance 62, 101729.
Sun, L., Abraham, S., 2021. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. J. Econometrics 225, 175–199.
Van Oordt, M., Zhou, C., 2019. Systemic risk and bank business models. J. Appl. Econometrics 34, 365–384.
Wang, Z., Wang, C., Fang, Z., 2023. Common institutional ownership and corporate misconduct. Manage. Decis. Econ. 44, 102–136.
Whited, T.M., Wu, G., 2006. Financial constraints risk. Rev. Financ. Stud. 19, 531–559.
Xu, X., Lu, Z., Tang, T., 2020. Method, measurement and inducement of leverage manipulation in Chinese listed companies. J. Manag. Sci. China 23, 1–26.
Zhu, R., Xin, X., Tan, K., 2024. Reverse mixed ownership reform: does state-owned capital injection inhibit corporate leverage manipulation? Financ. Res. Lett. 59, 104764.